MILITARY SPECIFICATION

RECORDER-REPRODUCER, SOUND, PORTABLE, MAGNETICALLY COATED TAPE

1. SCOPE

1.1 This specification covers a portable, dual track, two speed, magnetic tape, sound recorder-reproducer for use primarily in personnel training and entertainment.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

W-T-61 - Tapes, Recording, Magnetic Oxide Coated, and Reels, Recording Tape.

MILITARY

MIL-J-641 - Jacks, Telephone, General Specification for.
MIL-P-642 - Plugs, Telephone.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for.
MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-I-17623 - Interference, Radio, Requirements, Methods and Limits (14 KC to 1000 MC) for Electric Office Machines, Printing and Lithographic Equipment.
MIL-F-19207 - Fuseholders, Extractor Post Type, Blown Fuse Indicating and Nonindicating.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Applications for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N. Y.)

3. REQUIREMENTS

3.1 Qualification. - The recorder-reproducer equipment furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Description. - The recorder-reproducer shall consist of a single unit made up of the items in 3.2.1 through 3.2.4.

3.2.1 Tape transport. - The tape transport assembly shall contain all the mechanical components for transporting the tape. It shall have mounted on it the tape guides, capstan, capstan idlers, motors, tape counter and all operational controls required for operation of the tape transport.

3.2.2 Electronics assembly. - The electronics assembly shall contain all the power supply, amplifier and oscillator components necessary to amplify the signal for recording and playback.

3.2.3 Case. - A sturdy case shall be provided as an enclosure for the recorder-reproducer (see 3.5.7).

3.2.4 Loundspeaker. - A loudspeaker shall be mounted internally in the case to reproduce the recorded signal.

3.3 General features. - The recorder-reproducer shall be in accordance with the following

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paragraphs of Specification MIL-I-983 in addition to the requirements specified herein: (Whenever a requirement of Specification MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.)

General information
Definitions
Materials, general
Stressed or vital parts
Substitution of materials (equal or superior)
Fungus-inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Arc-resistant materials
Toxic materials
Lubricants and lubrication
Painting
Bolts, machine screws, studs and nuts
Locking devices
Washers
Parts - electrical - general
Electron tube or capacitor sockets
Capacitors
Variable resistors
Electrical tapes
Switches
Indicator lights and lampholders
Fuses
Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Temperature and humidity
Accelerated life (see 3.7.3)
Vibration and inclination (non-vital)
Primary power supply circuits
Power supply tolerances
Personnel protection
Shielding
Ground potential and grounding
Soldering
Wiring
Color coding
Electrical insulation
Drawings - general
Drawings - preliminary
Drawings - working
Schematic diagrams
Wiring diagram
Drawing list
Assembly drawings
List of onboard repair parts
Drawings - manufacturing
Interchangeability and standardization
Technical manuals
Repair parts
Repair parts (electronic)
Designation and marking (serial numbers are required)
Design for production
Nomenclature
Workmanship and general examination

3.4 Functional requirements. -

3.4.1 Time of operation. - The equipment shall be designed to operate continuously.

3.4.2 Recording and reproducing medium. - The recording and reproducing medium shall be 1/4 inch magnetically coated tape wound on a 7-inch diameter plastic reel. The equipment shall meet all requirements of this specification with 1 mil polyester base tape and 1 1/2 mil polyester or acetate base tape. The tape and reels shall be in accordance with Specification W-T-61.

3.4.3 Tape speeds. - Tape speeds of 3.75 and 7.5 inches per second (i.p.s.) shall be provided. The tape speed shall not vary from 3.75 or 7.5 i.p.s. by more than ± 2 percent while recording and reproducing in the middle portion of a reel of tape. The tape speed shall not vary from 3.75 or 7.5 i.p.s. by more than ± 5 percent during the first and last 10 percent of the tape length.

3.4.4 Interchangeability of recordings. - Recordings shall be interchangeable between all equipments manufactured in accordance with this specification. Track dimensions, clearances and relative directions of motion and recording shall apply and shall be in accordance with figure 1. Dimensions shall be considered nominal.

3.4.4.1 Azimuth adjustment. - An azimuth adjustment shall be provided for the magnetic heads. The heads shall be adjusted for maximum output using a standard equalization test tape or a Government approved method.

3.4.5 Erasing. - The equipment shall provide a means of erasing the recording by a high frequency erase method. Erasing shall automatically occur during the recording process in such a manner that the tape is erased prior to reaching the recording head. Erasure shall only occur when the recorder-reproducer is in the record mode.

3.4.6 Tape motion controls. - All controls essential to the operation of the tape transport shall be located on the same panel and area of the equipment. These shall include the controls in 3.4.6.1 through 3.4.6.4.

3.4.6.1 Play switch. - A switch (type can be contractor's option) shall be provided to bring the tape up to standard speed (3.75 or 7.5 i.p.s.) from the stop position in not more 1 second.

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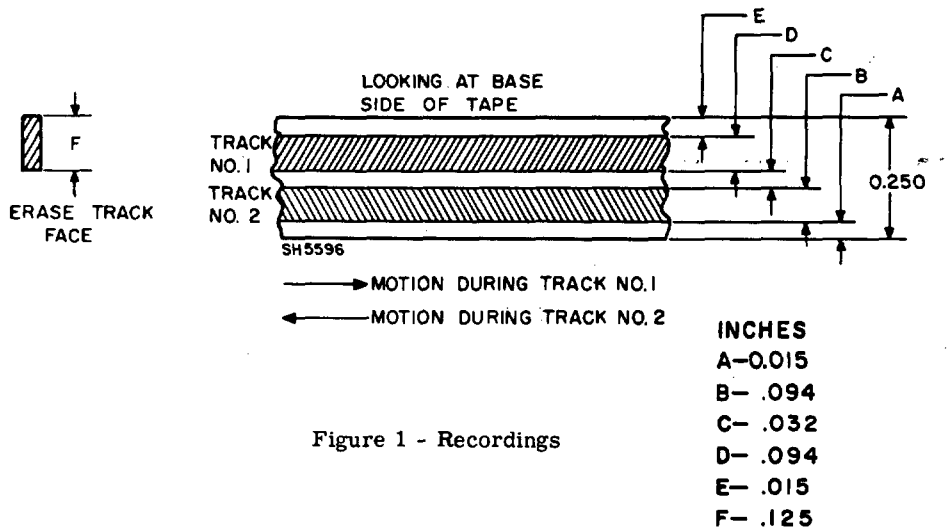


Figure 1 - Recordings

3.4.6.2 Record switch. - A switch (type can be contractor's option) shall be provided to bring the tape up to standard speed (3.75 or 7.5 i.p.s.) from the stop position in not more than 1 second.

3.4.6.2.1 Lockout switch. - An interlock switch shall be provided near the "Record" switch such that accidental record operation is impossible. It shall be necessary to have double function operation to place the equipment in the record mode.

3.4.6.2.2 When changing from the record mode to the playback mode, and vica versa, the recorder-reproducer shall go through the stop position.

3.4.6.3 Stop switch. - A switch (type can be contractor's option) shall be provided to bring the tape to a positive stop from its operating mode within 1 second. All rollers, pressure pads, etc., included in the tape path shall be disengaged in the stop position and the equipment shall be ready for threading.

3.4.6.4 Rewind and fast forward switch. - A switch (type can be contractor's option) shall be provided to completely wind (fast forward) and rewind a 1200 feet reel of tape in not more than 150 seconds. Maximum speed of the tape transfer shall result in no detectable deterioration of the the tape nor shall 1000 such operations result in a detectable deterioration of the equipment.

3.4.7 Tape speed change control. - Capstan speeds shall be selectable by a two position switch so as to provide the two tape speeds. In addition, selection of the proper equalization circuit for each speed shall be made automatically when the speed change switch is operated. This control need not be on the same panel as the controls of 3.4.6 but shall be visible from the front of the equipment.

3.4.8 Record-reproduce volume control. - A rotary type control shall be provided to adjust the gain of the amplifier while recording and reproducing. This control need not be included on the same panel as the controls of 3.4.6 but shall be visible from above the equipment.

3.4.9 Record level indicator. - A record level indicator shall be provided to indicate test record level. A dual neon lamp system may be used. If neon lamps are used, one shall begin to indicate 6 db below test record level and the other shall begin to indicate at the test record level. If volume unit (V.U.) meters are used, then zero

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db shall be at 6 db below test record level. This indicator shall be visible from above the equipment while in its operating mode.

3.4.10 Tape counter. - A veeder-Root, or equal, type counter shall be connected to either the supply or take-up reel to count the number of turns of the reel during operation.

3.4.11 Power cord. - An eight foot power cord shall be permanently attached to the recorder-reproducer. The cord shall be 3-conductor Underwriters Laboratory type SO or ST.

3.4.12 Attachment cord. - A five foot rubber covered cord fitted with alligator clips on one end and a type PJ-055B plug conforming to Specification MIL-P-642 at the other end shall be supplied with the equipment.

3.4.13 Fuses. - All fuses necessary for adequate protection of the recorder-reproducer shall be mounted on the front panel so that they may be easily replaced. Both lines of the 115 V AC supply shall be fused. Fuseholders shall be in accordance with Specification MIL-F-19207.

3.4.14 Power switch. - A power switch shall be provided which will disconnect both sides of the power supply line when in the "OFF" position. A pilot light shall be provided to indicate when the power switch is in the "ON" position.

3.4.15 Flutter and wow. - The amount of flutter and wow in the tape drive system shall not exceed 0.3 percent R.M.S. This requirement shall be met for both tape speeds during the recording and reproducing processes.

3.4.16 Microphone and cord. - A microphone shall be supplied which is either a dynamic or magnetic type. When a 10 dyne per square centimeter signal of 1000 c.p.s. is applied to the face of the microphone at least 20 milivolts shall be developed across the microphone input of the recorder. The microphone case shall be designed for desk mounting. This may be accomplished by supplying a desk stand. If the microphone is supplied without a desk stand, it shall have a base area of at least 3 square inches. The microphone shall be provided with an integral stand adapter having a 5/8 inch - 27 thread which will allow the microphone to be mounted on a conventional floor stand. The microphone shall be supplied with a 10 foot rubber covered cord complete with a type PJ-055B plug in accordance with Specification MIL-P-642.

3.5 Mechanical requirements. -

3.5.1 Size and weight. -

3.5.1.1 Size. - Overall dimensions shall be held to a minimum. The sum of the height, width and length dimensions shall not exceed 35 inches, including the cover (when closed), but excluding mounting feet and carrying handle.

3.5.1.2 Weight. - The weight of the complete equipment, excluding accessories, shall not exceed 30 pounds.

3.5.2 Equipment assembly. - The recorder-reproducer shall consist of two main assemblies as follows:

(a) A tape drive transport and electronic assembly. The tape drive and electronic assembly shall be removable from the enclosure.
(b) An enclosure (external or integral to the recorder).

3.5.2.1 Tape transport. - The tape transport shall be designed to operate in the horizontal position. All elements required to move the tape from one reel to another shall be included in the tape drive system. During recording and reproducing, the tape speed shall be essentially a sole function of the peripheral velocity of the capstan surface. The surface of the capstan itself shall be extremely resistant to corrosion from the effects of humidity as might be encountered in the operation of the equipment in shipboard applications.

3.5.2.2 Tape threading. - The tape transport shall be designed to facilitate simple threading of the tape. Idler pulleys, tape guides and the like shall not have to be manipulated to accomplish the threading operation. There shall be no sharp corners or edges on or about the surface of the tape transport on which the tape might catch and tear, or which might result in an operational hazard for the operator. Tape threading operations shall be accomplished in a simple straight forward manner, with a minimum of opportunities for the unskilled operator to misinterpret the threading procedure. An outline of the threading path shall be provided on the front panel.

3.5.3 Tape handling. - At no time during the tape start, running or stop modes shall loops (2 inches in total length) or other unfavorable tape conditions develop.

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3.5.4 Ventilation. - The equipment shall be ventilated in such a manner that under any continuous normal operating condition no part or assembly shall attain a temperature which will tend to damage or to reduce the normal useful life. It is recommended that forced air colling by a fan blade driven by the motor be provided in a manner that will draw air in through the bottom of the equipment as it is set up for normal operation and expel the heated air through the top or sides.

3.5.5 Temperature rise. - Continuous operation of the recorder-reproducer at rated power for a period of 2 hours shall not produce a temperature rise of any part of the equipment, exclusive of electron tubes, motors, and resistors, in excess of 50°F. above the ambient when tested as specified in 4.3.16.

3.5.6 Roller deformation. - If the design of the equipment is such that it is possible to be in the "OFF" position (no power being supplied) with rubber rollers engaged, the rollers shall be of such a composition that they will not be permanently deformed to the extent that performance will be seriously impaired. If applicable, the equipment shall be subjected to the test specified in 4.3.18.

3.5.7 Case. - A case shall be provided as an enclosure for the recorder-reproducer. This case may be an integral part of the equipment. Strong, corrosion-resisting hardware shall be used and all external corners shall be rounded. If the case is fabricated of plastic, it shall be of the high shock resistant type. If the case is fabricated of wood, it shall be tongue-in-grove construction. The case shall be strong enough to resist the normal shocks and stresses that occur in frequent moving of the equipment from one location to another aboard ship. Removal of a single cover shall be the only opening necessary to set the equipment up for operation. The cover shall be secured to the case with heavy duty separable hinges (to permit removal of the cover from the enclosure) and latches sufficiently rugged to provide trouble free operation during the life of the equipment. Rubber mounting feet shall be provided on the bottom of the enclosure and shall mount by through-bolting.

3.5.7.1 Handle. - A carrying handle of adequate strength to withstand rough handling during transportation of the equipment shall be provided. The handle shall be located on the shortest dimension and shall be parallel to the longest dimension.

3.5.8 Accessory storage. - Storage facilities shall be provided in the carrying case for the following accessories which shall be provided with each equipment:

(a) One microphone complete with cord and plugs (see 3.4.16)
(b) One power cord (see 3.4.11)
(c) One attachment cord (see 3.4.12)

3.5.8.1 The following shall be supplied with the equipment, but storage facilities in the carrying case need not be provided:

(a) Two reels of tape conforming to type TAT1200AF250NA of Specification W-T-61, packed in fiberboard containers.

(b) One empty reel conforming to type ER250NA of Specification W-T-61.

3.6 Electrical requirements. -

3.6.1 Electrical components. -

3.6.1.1 Tubes and transformers. - The electron tubes and transformers used in the equipment shall be of a commercial grade.

3.6.1.2 Motors. - Requirements for materials included in this specification do not apply to electric motors, but the motors shall meet the temperature, humidity and all other requirements of this specification. The motors used in this equipment shall be AC motors.

3.6.1.3 Loudspeaker. - A permanent magnet type loudspeaker shall be provided as an integral part of each recorder-reproducer. The loudspeaker shall have a nominal impedance of 3.2 ohms when measured at 400 c.p.s. The output level of the loudspeaker, when measured 2 feet on axis, shall be at least 91 db relative to 0.0002 dynes per square contimeter at a frequency of 400 cycles with an input of 2 watts to the loudspeaker voice coil. The loudspeaker shall have a frequency response of plus or minus 7 db relative to a 1000 cycle signal between the limits of 200 and 5000 cycles when the output level is measured as specified in 4.3.2.

3.6.1.4 Pushbutton assembly switch. - A pushbutton assembly switch may be used is this equipment to perform all or part of the functions of the tape motion controls and the electrical controls as specified herein. Each section of the

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pushbutton assembly switch shall be capable of withstanding 200,000 actuations without degradation in performance when tested as specified in 4.3.17.2. After 200,000 actuations the switch shall be installed in the recorder-reproducer and the equipment shall be tested for conformance with the operational test specified in 4.3.17.2.

3.6.2 Amplifiers. -

3.6.2.1 Test record level. - The test record level is defined as the record head current at which the overall (input to output) total R.M.S. harmonic distortion does not exceed 5 percent when measured on a 1000 cycle tone when reproduced through the equipment at 2 watts output. A standard test record level tape will be supplied to the contractor by the Government. This standard tape shall be used in the following manner: The standard tape shall be played-back on the equipment with control knobs adjusted to give 2 watts at a distortion level of less than 5 percent. Another tape (not the standard tape) shall then be recorded with a particular input level. This latter tape shall then be reproduced with the controls set in the same position as they were when using the standard tape. The output shall then be at least 2 watts with distortion at, or less than, 5 percent.

3.6.2.2 Power supply. - The equipment shall operate in conformance with this specification from a normal power source of 115 volts, 60 cycle, AC and 50 cycle AC when specified (see 6.1). The equipment shall also be capable of recording and reproducing a useable recording within the power supply tolerances specified in Specification MIL-I-983.

3.6.2.3 Amplifier inputs. - The record amplifier shall have two inputs. One shall be a microphone input and the other shall be a bridging input.

3.6.2.3.1 Microphone input. - The microphone input impedance shall be 500,000 ohms ± 10 percent. The input jack shall be in accordance with Specification MIL-J-641 and shall be designed to receive a type PJ-055B plug in accordance with Specification MIL-P-642. The equipment shall operate satisfactorily when input voltages between 2 millivolts and 100 millivolts are applied to the microphone input.

3.6.2.3.2 Bridging input. - An input shall be provided having an impedance of 20,000 ohms ± 10 percent and shall operate satisfactorily from an input line of any impedance which will produce a minimum voltage of 200 millivolts or a maximum of 10 volts across the input. The input jack shall be in accordance with Specification MIL-J-641 and shall be designed to receive a type PJ-055B plug in accordance with Specification MIL-P-642.

3.6.2.4 Loudspeaker output. - An output jack shall be provided to operate an external loudspeaker having an impedance of 6 to 8 ohms. The jack shall be in accordance with Specification MIL-J-641 and shall be designed to receive a type PJ-055B plug in accordance with Specification MIL-P-642. The internal loudspeaker shall be disconnected when the plug is inserted.

3.6.2.5 Output power and distortion. - The output power from the recorder, using a tape recorded on this equipment, shall be at least 2 watts at not more than 5 percent harmonic distortion at 1000 cycles per second and shall not exceed 5 percent distortion over a range of 100 to 5000 c.p.s. at 3.75 inches per second and a range of 100 to 7500 c.p.s. at 7.5 inches per second when tested as specified in 4.3.9.

3.6.2.6 Equalization. -

3.6.2.6.1 Standard equalization tapes shall be used for testing conformance with the equalization requirements of this specification for both speeds (see 6.4). A pre-emphasis of approximately 8 db at 5000 c.p.s. relative to 0 db at 1000 c.p.s. is required to obtain the standard equalization characteristic. All low frequency equalization below 400 c.p.s. shall be accomplished in the play back amplifier.

3.6.2.6.2 The frequency response signals prerecorded on the standard equalization tapes shall be reproduced through the recorder-reproducer and shall meet the overall frequency response limits of 3.6.2.7 when tested as specified in 4.3.10.

3.6.2.7 Frequency response. - The frequency response of the recorder-reproducer (from input to the record amplifier to the output of the playback amplifier) shall be as follows:

Tape speed (i.p.s.)	Response
3-3/4	± 4 db at 100 c.p.s. to 5000 c.p.s.
7-1/2	± 4 db at 100 c.p.s. to 7500 c.p.s.

3.6.2.8 Signal to noise ratio. - The signal to overall noise ratio shall be at least 35 db when the signal level is reproduced at not greater than 5 percent harmonic distortion and measured against the noise level in accordance with 4.3.12.

3.6.2.9 Radio interference. - The equipment shall conform to the requirements for portable electric equipment of Specification MIL-I-17623 when tested as specified therein.

3.7 Physical durability. -

3.7.1 Exposure to extreme temperatures. - The equipment shall not suffer any permanent degradation in performance as the result of exposure to temperatures of from minus 40°F. to plus 160°F.

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3.7.2 Resistance to temperature and humidity. - Temperature and humidity cycling, as specified in 4.3.15, shall not render the equipment inoperative, produce corrosion or cause a permanent degradation in performance.

3.7.3 Accelerated life. - The recorder-reproducer shall be capable of 144 hours continuous operation when tested in accordance with 4.3.17.

3.7.4 Rough handling. - All parts of the equipment shall be designed to have no critical adjustments such that normal rough handling and shipment will necessitate readjustment. When packed as required in the specific contract or order, the recorder-reproducer shall not be damaged by dropping when tested as specified in 4.3.19.

3.8 Painting. - The painting requirements of Specification MIL-I-983 shall apply only to exposed metallic surfaces when the carrying case cover is removed.

3.9 Designation and marking. - In addition to the designation and marking requirements of Specification MIL-I-983, all controls and jacks shall be permanently marked to indicate function.

3.10 Technical manuals. - Type IV technical manuals shall be furnished in accordance with Specification MIL-M-15071. The manuals shall include operating and maintenance instructions. Working drawings in accordance with Specification MIL-I-983 (see 3.3) shall be included in the manuals.

3.11 Maintenance and servicing. -

3.11.1 Removal from case. - The removal of the tape transport and electronic sections from the case shall be accomplished easily and straightforwardly. Removal of screws shall be the only necessary procedure for complete removal of the transport and electronic assembly from the enclosure.

3.11.2 Replacement of components. - All electrical and mechanical components shall be easily removable for servicing and replacement. No involved mechanical adjustment techniques involving belts, rollers, pads, etc., shall be required before proper operation is restored. A minimum of disassembly shall be required for replacement of electron tubes.

3.11.3 Lubrication. - All bearing surfaces shall be preferably permanently sealed. If the bearings are not sealed, they shall not require lubrication more often than once every 30 days of continuous, 8 hours per day operation. Excess lubrication shall not result in deterioration of equipment, or jeopardize in any way the correct functioning of the recorder.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General. - The sampling procedures and the inspection to be performed on the equipment shall be as specified herein (the attention of the contractor is invited to the applicable paragraphs of Specification MIL-I-983 for information concerning inspection such as where conducted, failures, test conditions, etc.).

4.2.1 Qualification. - Qualification inspection shall be conducted in a laboratory satisfactory to the Bureau of Ships [1]. One recorder-reproducer shall be subjected to the following examination and tests in the order listed:

Test	Requirement Paragraph	Test Paragraph
General inspection and operating test	3.3	4.3.1
Supply line voltage and frequency	3.3	MIL-I-983
Radio frequency noise interference	3.6.2.9	MIL-I-17623
Inclination	3.3	MIL-I-983
Tape speed	3.4.3	4.3.4
Rewind and fast foward speed	3.4.6.4	4.3.5
Record level indicator	3.4.9	4.3.6
Flutter and wow	3.4.15	4.3.13
Microphone	3.4.16	4.3.3
Dimensions	3.5.1	4.3.8
Temperature rise	3.5.5	4.3.16
Roller deformation	3.5.6	4.3.18
Loudspeaker	3.6.1.3	4.3.2
Input impedance	3.6.2.3	4.3.7
Output power and distortion	3.6.2.5	4.3.9
Equalization	3.6.2.6	4.3.10

[1] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.3).

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Test	Requirement Paragraph	Test Paragraph
Frequency response	3.6.2.7	4.3.11
Signal to noise ratio	3.6.2.8	4.3.12
Pushbutton assembly switch	3.6.1.3	4.3.17.2
Exposure to extreme temperatures	3.7.1	4.3.14
Resistance to temperature and humidity	3.7.2	4.3.15
Accelerated life	3.7.3	4.3.17
Drop test	3.7.4	4.3.19
Vibration (non-vital)	3.3	MIL-I-983

4.2.2 Acceptance inspection. - The equipment selected in accordance with 4.2.3 shall be subjected to the following examination and tests in the order listed:

Group A	Requirement Paragraph	Test Paragraph
General inspection and operating test	3.3	4.3.1

Group B	Requirement Paragraph	Test Paragraph
Tape speed	3.4.3	4.3.4
Rewind and fast forward speed	3.4.6.4	4.3.5
Record level indicator	3.4.9	4.3.6
Flutter and wow	3.4.15	4.3.13
Equalization	3.6.2.6	4.3.10
Output power and distortion	3.6.2.5	4.3.9
Frequency response	3.6.2.7	4.3.11
Signal to noise ratio	3.6.2.8	4.3.12

Group C	Requirement Paragraph	Test Paragraph
Radio frequency noise interference	3.6.2.9	MIL-I-17623
Resistance to temperature and humidity	3.7.2	4.3.15
Inclination	3.3	MIL-I-983
Vibration (non-vital)	3.3	MIL-I-983
Accelerated life	3.7.3	4.3.17
Temperature rise	3.5.5	4.3.16
Dimensions	3.5.1	4.3.8
Roller deformation	3.5.6	4.3.18
Drop test	3.7.4	4.3.19

4.2.3 Sampling for acceptance inspection. -

4.2.3.1 Inspection lot. - All equipment presented for acceptance inspection at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.2.3.2 Sampling for group A inspection. - All equipment shall subjected to the group A inspection listed in 4.2.2.

4.2.3.3 Sampling for group B inspection. - A sample of equipments shall be selected from each inspection lot in accordance with table I and shall be subjected to each of the group B tests listed in 4.2.2 with lot acceptance based on sampling requirements of Standard MIL-STD-105. Roller deformation (normally a group C test) shall be performed on one equipment per contract.

Table I. - Sampling for group B inspection.

Number of equipments in inspection lot	Number of equipments in sample	Number of equipments nonconforming on any group B test	
		Acceptance number	Rejection number
3 and under	All	-	-
4 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4.2.3.4 Sampling for group C inspection. - Group C inspection will be required when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be subjected to the group C inspection listed in 4.2.2 with acceptance based on 4.2.3.6.

4.2.3.5 Acceptance - group A and B inspection. - The results of each test shall be compared with this specification. Failure to conform to this specification for any test shall be counted as a defect and the equipment shall be rejected. If the number of nonconforming equipments in any group B sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.2.3.6 Acceptance - group C inspection. - The results of each test shall be compared with this specification. In the event of failure to conform to this specification for any group C test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

4.3 Procedures. -

4.3.1 General inspection and operating test. - The general inspection shall be conducted in

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accordance with Specification MIL-I-983. The operating test shall consist of the following: Voice or other suitable signals shall be recorded through the microphone and bridging inputs. The microphone supplied with the equipment shall be used for recording through the microphone input. The accessory cord supplied with the equipment shall be used for recording through the bridging input. The signals shall then be reproduced alternately through the internal loudspeaker and an external loudspeaker connected to the equipment through the external speaker jack. All electrical and mechanical controls shall be actuated during this test to determine functional conformance with the requirements of this specification.

4.3.2 Loudspeaker. - The loudspeaker removed from the recorder shall be set up under free-field conditions and tested as follows:

4.3.2.1 Output level. - A 400 cycle signal of 2 watts shall be applied to the voice coil of the loudspeaker. The output level shall be measured 2 feet on axis, when the speaker is mounted in a simulated infinite baffle to determine conformance with 3.6.1.3.

4.3.2.2 Frequency response. - Signals between 200 and 5000 cycles shall be applied to the voice coil of the loudspeaker with an input of 1 watt while the loudspeaker is mounted in a simulated infinite baffle. Measurements shall be taken 2 feet on axis to determine conformance with 3.6.1.3.

4.3.3 Microphone. - The sound source for this test shall be the Navy standard artificial mouth. The microphone shall be set up for test under free-field conditions at a point on the axis 3/4 inch in front of the ring on the artificial mouth. The microphone shall be connected to a 500,000 ohm resistive load. The artificial mouth shall be adjusted to deliver a sound pressure of 10 dynes per square centimeter of 1000 c.p.s. at the face of the microphone. The voltage developed across the resistor shall be measured to determine conformance with 3.4.16.

4.3.4 Tape speed accuracy. - A 450 inch length of 1/4 inch recording tape in the center portion of a 1200 foot reel of tape shall be marked at each end by splicing tape or other suitable means. A 450 inch length shall also be marked off at the beginning and end of the reel of tape. The tape shall then be run through the equipment at the 3.75 inch per second speed. The time intervals for the measured sections to pass a given point on the equipment shall be determined. The average tape speed shall then be calculated to determine conformance with 3.4.3. The same procedure shall be followed for the 7.5 inch per second speed.

4.3.5 Rewind and fast forward speed. - A 1200 foot reel of tape shall be completely run through the equipment at the fast forward and rewind speeds to determine conformance with 3.4.6.4.

4.3.6 Record level indicator setting. - A 20 millivolt signal of 1000 cycles shall be applied to the microphone input and the record-reproduce volume control shall be adjusted to the test record level. If an electron eye tube is used, the record-reproduce volume control shall then be adjusted for an attenuation of 3 db and the electron eye shall be observed to see if the eye remains closed. The record-reproduce volume control shall be adjusted to give an additional attenuation of 2 db and the electron eye shall be observed to see that the eye opens a noticeable amount. If neon lights are used, the lights shall be observed to see if both lights are indicating when the record-reproduce volume control is adjusted for test record level. The record-reproduce volume control shall be adjusted to give an additional attenuation of 6 db and the lights observed to see if one light is indicating. The record-reproduce volume control shall be adjusted to give 2 db additional attenuation and the lights observed to see that neither light is indicating.

4.3.7 Input impedance. - Input impedance shall be measured at 1000 c.p.s. for conformance with 3.6.2.3.

4.3.8 Dimensions. - The dimensional parameters shall be measured for conformance with 3.5.1.

4.3.9 Output power and distortion. - Distortion shall be measured using a signal generator having an output distortion content of less than 0.3 percent. Input signals of 2 millivolts and 200 millivolts shall be recorded through the microphone and bridging inputs respectively at the test record level for frequencies of 100 and 1000 c.p.s. With the same input voltages the record-reproduce volume control shall be adjusted to give a recording level 6 db below the test record level. A 3000 cycle signal shall then be recorded. The record-reproduce volume control shall then be adjusted to give a record level 12 db below the test record level and a signal of 5000 cycles shall be recorded. The tape shall then be played back with the record-reproduce volume control adjusted to give at least 2 watts output for the 1000 cycle signal. The tone control shall be adjusted for no attenuation. Distortion measurements for all the recorded signals shall then be measured at this volume and tone control setting across a 4 ohm resistive load to determine conformance with 3.6.2.5. The test shall be repeated for inputs of 100 millivolts and 10 volts to the microphone and bridging inputs respectively at the 7.5 inch per second speed, distortion measurements shall be made as above

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and, in addition, a measurement shall be made at 7500 c.p.s.

4.3.10 Equalization. - The 100, 400, 1000 and 5000 c.p.s. prerecorded signals on the standard equalization test tape shall be reproduced on this equipment with tone control set for no attenuation. Measurements shall be made across the loudspeaker output with power being dissipated into a 4 ohm resistive load to determine conformance with 3.6.2.6 at the 7.5 inch per second speed, equalization shall be measured as above with the 7.5 inch per second equalization tape.

4.3.11 Frequency response. - Tests shall be conducted with input signals of 20 millivolts and 500 millivolts applied to the microphone and bridging inputs, respectively. The record-reproduce volume control shall be set for that level which is 15 db below test record level. Signals shall be recorded on the tape at various frequencies within the frequency response limits specified in 3.6.2.7. The signals shall be played back with controls set so that a 1000 cycle per second signal will be 15 db below rated output of the playback amplifier. The response shall be measured across a 4 ohm resistive load to determine conformance with 3.6.2.7. This procedure applies to both speeds.

4.3.12 Signal to noise ratio. - A 1000 cycle signal shall be recorded on both tracks at test record level. The signal shall then be played back with the reproduce volume adjusted to give an output of 2 watts containing not more than 5 percent harmonic distortion across a 4 ohm resistive load. One track shall then be rerecorded with no signal input. The erased portion shall then be played back with the record-reproduce volume control set at the same position as when the recorded signal was played back. The noise level shall then be measured across a 4 ohm resistive load to determine conformance with 3.6.2.8.

4.3.13 Flutter and wow. - The recorder-reproducer equipment shall be tested for conformance with 3.4.15 by using a Furst meter or any other method satisfactory to the Bureau or agency concerned. This test shall be run at both speeds.

4.3.14 Exposure to extreme temperatures. - The equipment, exclusive of recording tape, shall be kept at minus 40°F. for 16 hours and then returned to plus 77° ± 9°F. for 4 hours. The equipment shall next be raised to plus 160°F. for 6 hours and then returned to plus 77° ± 9°F. for 2 hours. The equipment shall then be given the operational test, output distortion, frequency response, signal to noise ratio and flutter and wow tests to determine conformance with 3.7.1.

4.3.15 Temperature and humidity cycling. - This test shall consist of recording and reproducing radio program or other suitable electrical signals while the equipment is subjected at least four times to the following temperature and humidity cycle. At least 6 hours of this test shall consist of recording. The equipment shall be subjected to a temperature of plus 90°F. with a relative humidity of 90 percent for a period of 18 hours. The equipment shall then be immediately subjected to a temperature of plus 40°F. with a relative humidity of 90 percent for a period of 6 hours. The hot and cold cycle shall immediately begin again for a total of 4 cycles. Immediately after the fourth cycle, the equipment shall be given an operational test to determine that it is functioning properly and it shall be examined for corrosion and other damage. A recheck of the operational test, output distortion, frequency response, signal to noise ratio and flutter and wow tests shall be made 24 hours after the fourth cycle to determine conformance with 3.7.2.

4.3.16 Temperature rise. - The equipment shall be operated continuously at line voltage of 115 volts and at an ambient temperature of 122°F. The equipment shall then be tested for conformance with 3.5.5.

4.3.17 Accelerated life. -

4.3.17.1 Recorder-reproducer. - The equipment shall be operated continuously for a period of 144 hours. The test shall consist of recording and reproducing radio program material or other suitable electrical signals. At least 60 hours of this test shall consist of recording. At least 2 hours of this test shall consist of completely winding and rewinding a 1200 foot reel of tape at the fast forward and rewind speeds. An endless loop tape may be used for a portion of this test not exceeding 96 hours. Upon completion of 144 hours operation, the equipment shall be retested for conformance with the operational test, frequency response and flutter and wow tests of this specification to determine conformance with 3.7.3.

4.3.17.2 Pushbutton assembly switch. - The switch from the recorder-reproducer or an identical one submitted by the contractor shall be tested while disconnected from the recorder-reproducer. Each switch of the pushbutton assembly shall be actuated 200,000 times. The assembly shall then be installed in the recorder-reproducer and the equipment shall be retested to determine conformance with 3.6.1.4.

4.3.18 Roller deformation. - The equipment shall be tested to determine conformance with 3.5.6. The equipment shall be allowed to stand

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for 96 hours with the rollers engaged. Then, with the rollers engaged, the equipment shall be operated for a 15 minute period. At the end of this time, the equipment shall be tested to determine conformance with 3.4.15.

4.3.19 Rough handling. - The recorder-reproducer packed for normal shipment shall be held 24 inches from a concrete floor and dropped as follows: (a) Once on a bottom corner; (b) Once on an opposite top corner; and (c) Once on the bottom face. The equipment shall then be unpacked and given the operational tests specified in 4.3.1 to determine conformance with 3.7.4.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of shipboard repair parts. -

5.1.1 Basic equipment. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Shipboard repair parts. - The shipboard repair parts shall be preserved and packaged level A; packed level C and marked levels A and C, respectively, in accordance with Specification MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1).

5.2.1.1 Preservation, packaging, packing and marking. - The equipment and accessories, repair parts and technical publications shall be preserved and packaged levels A or C; packed levels A or B, as specified, and marked in accordance with Specification MIL-E-17555.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Quantity of technical manuals required (see 3.3 and 3.10)
(c) Repair parts required (see 3.3)
(d) When required to operate from a 50 cycle AC power source (see 3.6.2.2).
(e) Preservation, packaging, packing or marking requirements other than these required by 5.1 (see 5.2).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-22717, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

6.3 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

6.4 Standard equalization and record test tape. - Standard equalization and record test tape will be made available to the contractor on a loan basis through the bureau or agency concerned.

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6.5 Nomenclature. - The name of parts should conform to Cataloging Handbook H6-1 published by the Armed Forces Supply Support Center.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH OBSOLETE PLATE NO. 15419

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